Getting Started with Python

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Download the Following Materials

- WinPython (Mac users: Anaconda)
- Get Notepad++
- Get git
- From the FH-library (you must be logged in!!), get the book Hands-on Signal Analysis with Python "SaPy"
- Personally, I also really like Directory Opus. But it is commercial, and slightly more expensive. Give it a try - you may like it too.

On Your Computer

- Make sure you have a program to zip and unzip data installed (e.g. `7zip`).
- Clean up your *Windows Desktop*: You should not have more than seven icons on it. And don't just pull files onto the Desktop, always save them into folders!
- Turn on *Windows File Explorer* | *View -> Show -> File Name Extensions*.
- Make sure you have a clear file-structure on your computer:
  - Know your home-directory! (in the `powershell`, you can get there with `~`; in the `command window` you see it when you type `%USERPROFILE%`)
  - Either from there, or from the `Documents` directory, make a sub-directory `FH`, and there a separate folder for each lecture.
- Go to https://work.thaslwanter.at/py_intro , Chapter *Installation* , and install a Python Distribution(e.g. WinPython) on your computer.
- Try to work through that chapter as far as possible. Note that there is a video at the bottom to help you.

Personalize your computer

The following steps are for Windows users. For Mac, check out the instructions in *SaPy*, chapters 2.1.2. (Installation of Python) and 2.3.3. (Personalizing IPython/Jupyter)

- Set Environment path so that Python gets detected (The following assumes you have installed WinPython in the folder `C:\Programs`!)
  - in Alt+Win | Edit environment variables for YOUR account | Path | Edit | New | `C:\Programs\WPy64-31470\python\Scripts` | OK | OK
  - the same, with `C:\Programs\WPy64-31470\python`
- Add `~\ipynb.bat` to your home directory `jupyter lab %1`
- Add `~\ipy.bat cd [here you have to enter the directory you want to work in] jupyter qtconsole`

The Terminal

- Try out *Windows Command Prompt* (`cmd`) or *Terminal* (`pwsh` or `terminal`). Know what `*` means (Wildcard: "any file/folder containing").
- You may also want to check out powershell-tips or cmd-tips.

Recommended Resources

- Get the Python package FreeSimpleGUI: `pip install FreeSimpleGUI` (in the terminal)
- Get PyCharm + Jetbrains StudentPack
- Get GitHub Education

Check your first Python Skills

- Is Python running on your computer?
- Where is the Python executable located?
- What is the difference between Python and IPython?
- How do you check if your program is running properly? (Answer: try to run it from the command line.)
- Do you know how to run an IPYNB-file?
- Do you know the most important keyboard shortcuts in JupyterLab?